

Parking Spot Detection Using LiDAR

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Introduction

- * Parking spot detection is a key feature for autonomous vehicles (AV) and advanced driver assistance systems (ADAS).
- * This project explores using LiDAR (Light Detection and Ranging) technology with machine learning models to detect parking spots, utilizing its ability to capture 3D spatial data for precise results in real time.

Background

- * Traditional parking detection methods, such as camera-based vision, ground sensors, or GPS, often struggle with challenges like poor lighting, occlusions, and environmental noise.
- * While SLAM (Simultaneous Localization and Mapping) can address this challenge, its computational complexity and high latency necessitate alternative approaches.
- * LiDAR's 3D mapping improves accuracy while avoiding SLAM's computational overhead.

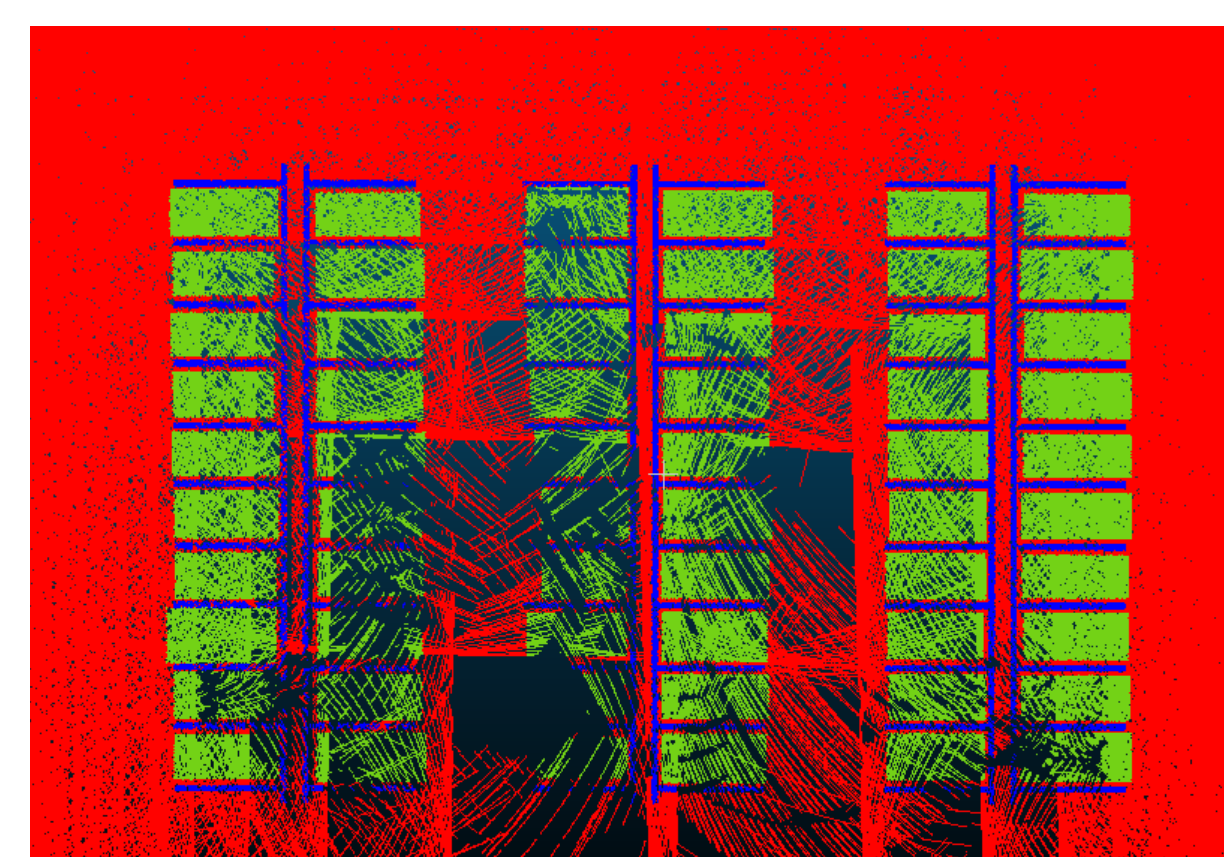


Parking Lot from CARLA Simulator

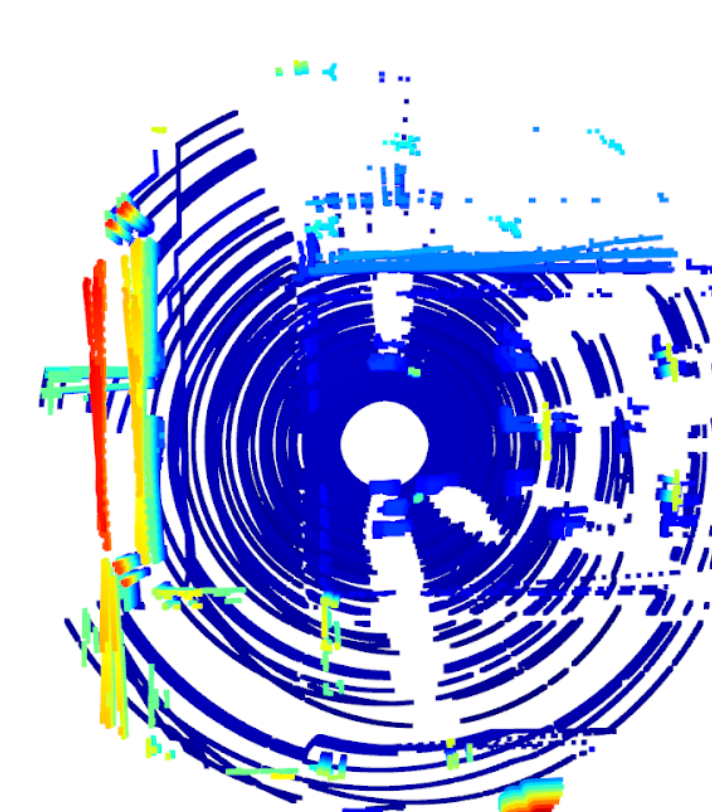
Methodology

Data Collection

- * We use the CARLA simulator as the primary environment for data collection and testing. A town is selected, and a base 3D point cloud map is generated as a reference.
- * Parking lot annotations are manually created for precise labeling, segmenting point clouds to detect parked vehicles and available spaces.

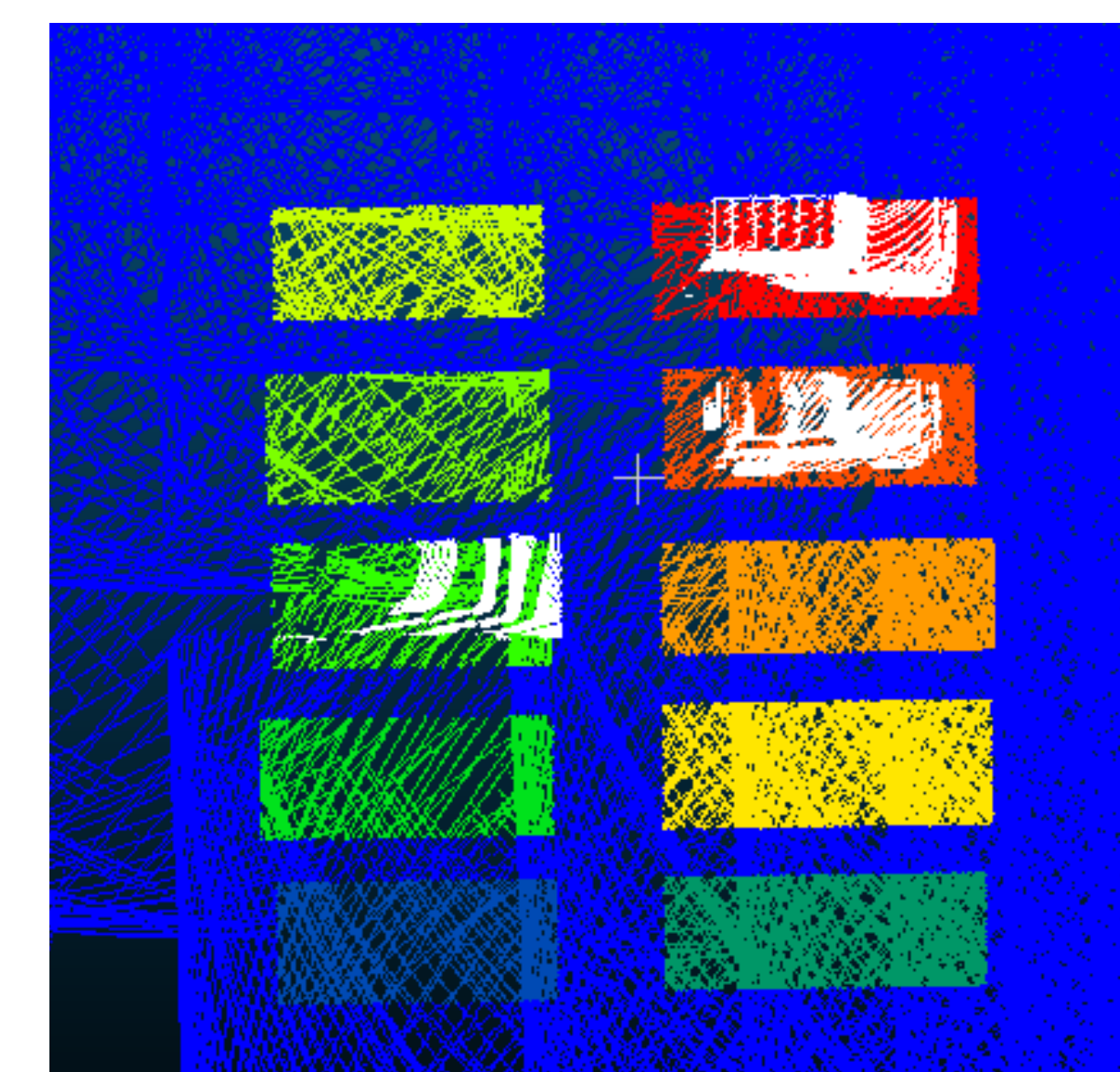
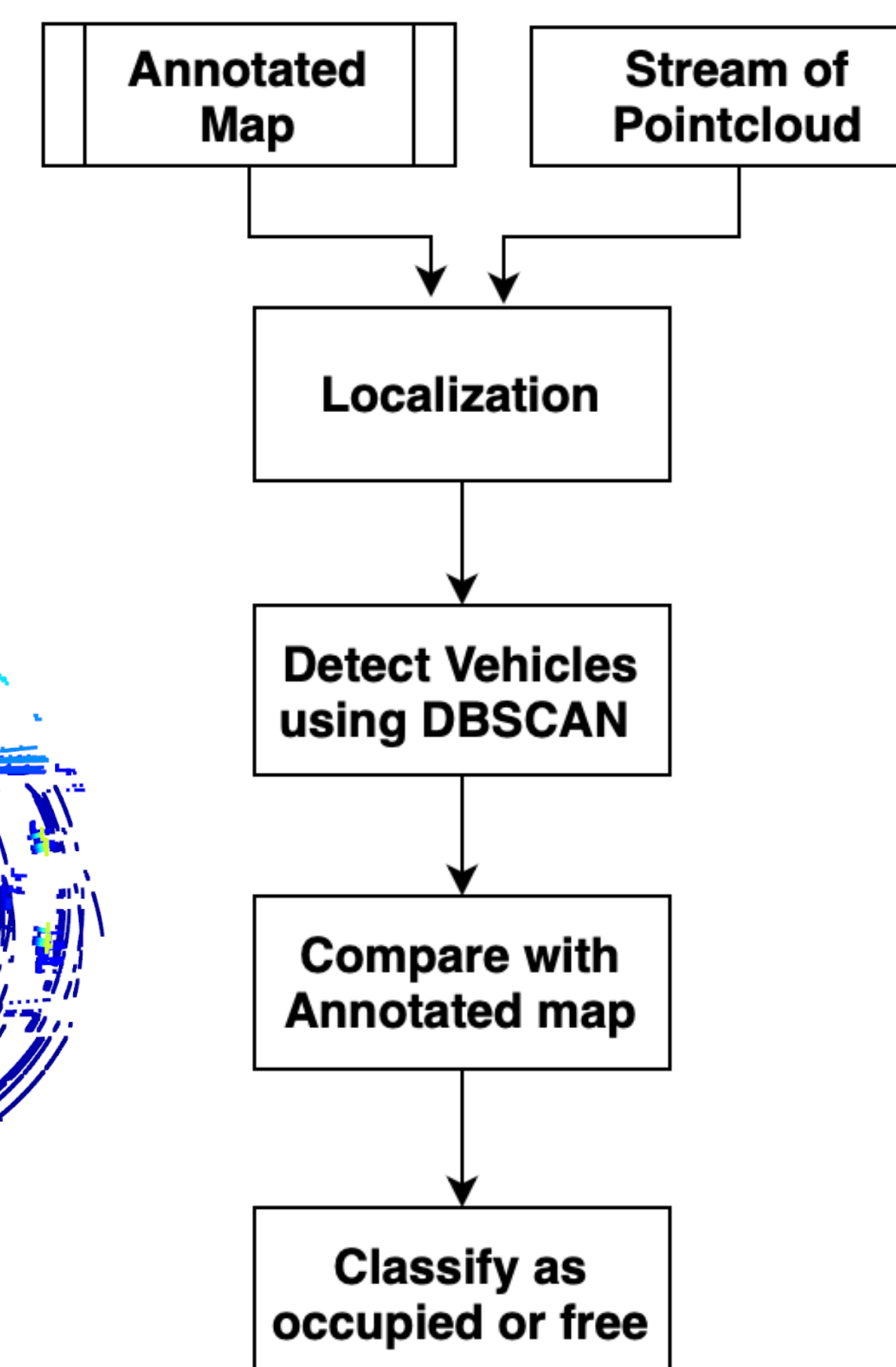


Segmented Parking Spaces
Green - spaces, Red - Ground points



LiDAR Trace

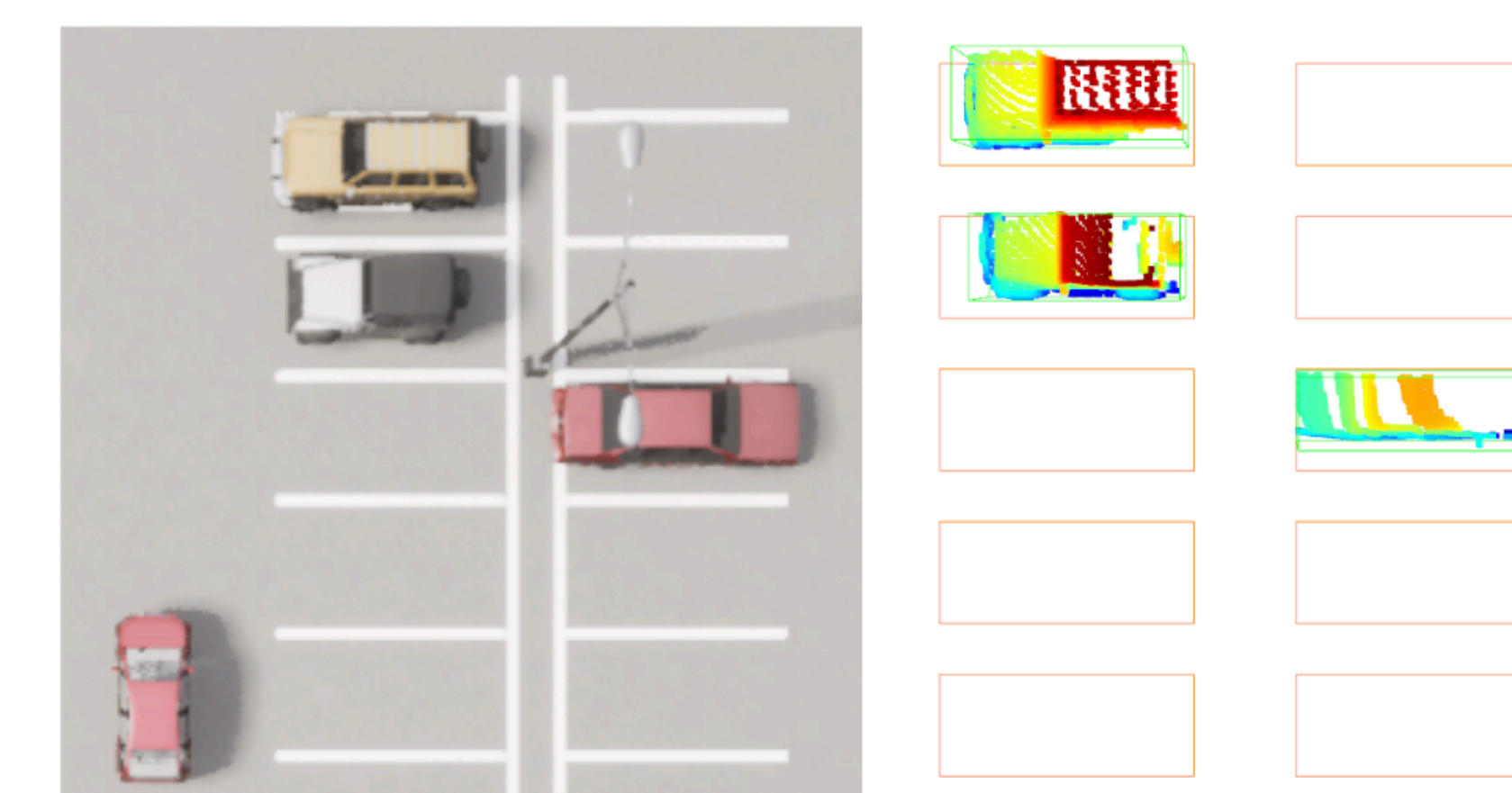
Workflow



Dynamic Map Output for 1 Frame

Results

The proposed system achieves higher detection accuracy than vision-only methods across varying environmental conditions. Its integration with AV parking assistance enhances automation and efficiency.



Vehicle clusters detected using DBSCAN algorithm

- * DBSCAN achieves 0.8 seconds per frame, making it ~40–60% faster than typical SLAM methods (~1.5–2.0 seconds per frame).
- * Achieved a +7% absolute accuracy gain over vision-only methods under low-light conditions, increasing detection rates from 82% to 89% on the test set (n = 100 frames).

Conclusion

- * DBSCAN offers faster and more robust clustering in low-light conditions compared to traditional methods.
- * Future Work: Validation on real-world datasets, crowdsourced data updates via connected vehicles.